

Computer Vision - 2026

Week #06. Segmentation and Object Detection

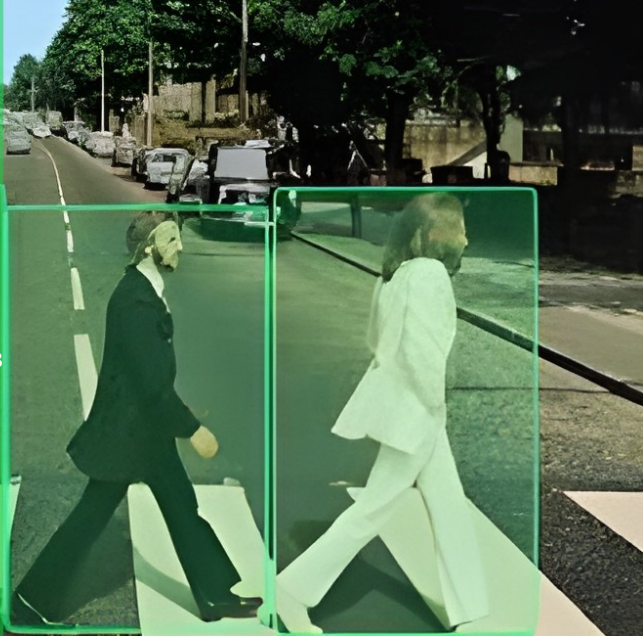
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Agenda

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Outcomes

Introduction

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- 2 Introduction
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Section 1. Outcomes

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By the end of this week, students will be able to:

- 1 **Formulate** segmentation and detection as structured prediction problems with precise mathematical notation.
- 2 **Explain** semantic segmentation architectures (U-Net, DeepLab) and the YOLO detection pipeline (backboneâneckâhead, loss, NMS).
- 3 **Implement** a YOLO detector and a semantic segmentation model, including data preparation, loss selection, and evaluation.
- 4 **Interpret metrics** (mIoU, Dice, etc.) and diagnose common failure modes.
- 5 **Select appropriate models** for real-time systems, robotics, medical imaging, or high-accuracy offline pipelines.

Key Takeaway

Segmentation and detection are structured vision problems requiring joint optimization of classification and spatial localization; model choice is a trade-off between accuracy, latency, and supervision cost.

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Section 2. Introduction

Types of Image Segmentation

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Segmentation Overview

Image segmentation is the process of partitioning an image into meaningful regions. It plays a key role in medical imaging, autonomous driving, and scene understanding.

- **Semantic Segmentation:** Classifies each pixel into predefined object categories.
- **Instance Segmentation:** Identifies individual object instances within categories.
- **Panoptic Segmentation:** Unifies semantic and instance segmentation by labeling both things (objects) and stuff (background regions).

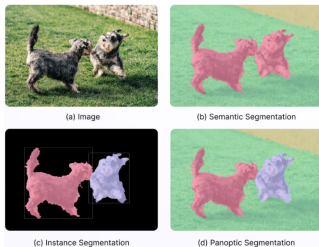


Figure: Types of segmentation problems.

Problem Formulation: Structured Outputs

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Input $x \in \mathbb{R}^{H \times W \times 3}$ (H, W : res., K : classes, θ : params, $u = (i, j)$: pixel, N : objects).

Semantic Segmentation

$f_{\theta}(x) = P \in [0, 1]^{H \times W \times K}$, $P(u, k) = \mathbb{P}(y_u = k \mid x)$.

Output $\hat{Y} \in \{1, \dots, K\}^{H \times W}$ via $\hat{y}_u = \arg \max_k P(u, k)$.

Object Detection

$f_{\theta}(x) = \{(b_i, c_i, s_i)\}_{i=1}^N$, box $b_i \in \mathbb{R}^4$, class c_i , score $s_i \in [0, 1]$.

Instance Segmentation

$f_{\theta}(x) = \{(m_i, c_i, s_i)\}_{i=1}^N$, mask $m_i \in \{0, 1\}^{H \times W}$, class c_i , score s_i .

Taxonomy: Tasks and Outputs

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Four Canonical Tasks

Task	Output	Supervision	Typical Models
Semantic Segm.	Dense per-pixel class map $H \times W$	Pixel labels (stuff + things)	FCN, U-Net, DeepLab, SegFormer
Instance Segm.	Set of masks $\{m_i\}$ + classes	Masks per instance (expensive)	Mask R-CNN, SOLO, Mask2Former
Panoptic Segm.	Single map with instance IDs + stuff	Stuff + instance masks (hardest)	Panoptic-FPN, Mask2Former, DETR
Object Detection	Set of boxes $\{b_i\}$ + classes	Boxes per instance (cheaper)	Faster R-CNN, YOLO, RetinaNet, DETR

Implementation View: Data and Annotation Pipelines

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Data Shapes

- **Detection dataset:** $\{(x^{(n)}, \{(b_i^{(n)}, c_i^{(n)})\}_{i=1}^{N_n})\}_{n=1}^M$
- **Segmentation dataset:** $\{(x^{(n)}, y^{(n)}), y^{(n)} \in \{1, \dots, K\}^{H \times W}\}_{n=1}^M$
- **Instance dataset:** $\{(x^{(n)}, \{(m_i^{(n)}, c_i^{(n)})\})\}_{n=1}^M$

Annotation Cost (Engineering Reality)

- Boxes: fast labeling, scalable to large datasets (COCO-like).
- Masks: slow labeling, often smaller datasets / domain-specific (medical, industrial QC).
- Panoptic: most expensive; usually limited data, heavy augmentation.

Losses: What We Optimize (Overview)

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Segmentation (dense)

- **Cross-Entropy (CE):** pixel-wise classification.
- **Dice / IoU losses:** robust for class imbalance and small structures.

$$\mathcal{L}_{seg} = \lambda_{CE} \mathcal{L}_{CE} + \lambda_{Dice} \mathcal{L}_{Dice}$$

Detection (set of objects)

- Classification loss: CE / focal loss (handles imbalance: many negatives).
- Box regression: SmoothL1, IoU/GIoU/DIoU/CIoU family.

$$\mathcal{L}_{det} = \mathcal{L}_{cls} + \lambda_{box} \mathcal{L}_{box} + \lambda_{obj} \mathcal{L}_{obj}$$

Metrics: What We Report (and Why It Matters)

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Segmentation Metrics

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$$IoU = \frac{|Y \cap \hat{Y}|}{|Y \cup \hat{Y}|}, \quad Dice = \frac{2|Y \cap \hat{Y}|}{|Y| + |\hat{Y}|}$$

- **mIoU**: mean IoU over classes (common in semantic segmentation).
- **Dice**: widely used in medical imaging (small targets).

Detection Metrics (COCO-style)

- **AP@0.5**: easier metric (Pascal VOC tradition).
- **AP@[0.5:0.95]**: strict metric; averages over IoU thresholds.
- **AR**: average recall; highlights missed objects.

Analytics: Typical Failure Modes by Task

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Structured Error Taxonomy

- **Segmentation:** boundary errors, small structures disappear, class imbalance, label noise, domain shift.
- **Detection:** localization error, confusion between similar classes, duplicates, missed small/occluded objects.
- **Instance/Panoptic:** overlapping instances, crowd scenes, thin objects, inconsistent instance IDs.

Symptom	Metric signature	Likely cause / fix
Miss small objects	low AR, AP_small	increase resolution, FPN, augment, re-weight samples
Too many duplicates Blurry boundaries	AP drops, many FP mIoU ok, Dice low	tune NMS / use end-to-end set models add boundary loss / higher-res decoder / better aug
Domain shift	all metrics drop	adaptation, better pretraining, prompt-able models (SAM)

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Section 3. Semantic Segmentation

Semantic Segmentation: Core Idea

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Goal

Predict a **dense** label map: each pixel $u = (i, j)$ gets a semantic class.

$$y \in \{1, \dots, K\}^{H \times W}, \quad f_{\theta}(x) = P \in [0, 1]^{H \times W \times K}$$

$$P(u, k) = \mathbb{P}(y_u = k \mid x), \quad \hat{y}_u = \arg \max_k P(u, k)$$

What makes it non-trivial?

- Need **global context** (class identity) + **local precision** (boundaries).
- Strong **class imbalance** (background dominates), small objects/structures.
- Output is **structured**: neighboring pixels are correlated.

Typical families

Encoder-decoder (U-Net), atrous/ASPP (DeepLab), transformer-based dense heads (SegFormer).

U-Net Architecture

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Conclusion

- U-Net consists of a contracting path (encoder) and an expansive path (decoder).
- Skip connections concatenate corresponding encoder and decoder feature maps to retain spatial details.
- Final layer applies a 1×1 convolution to obtain pixel-wise classification.

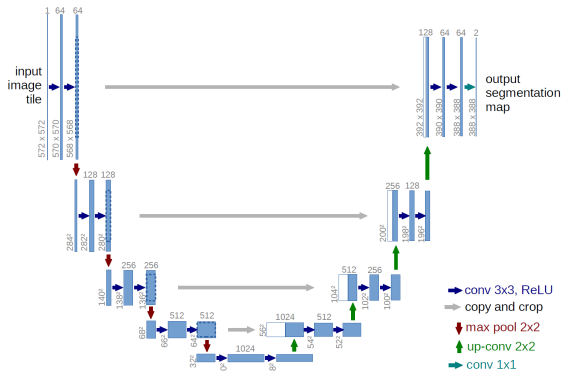


Figure: Detailed U-Net structure [Ronneberger et al., 2015].

Residual Connections in U-Net

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Key Idea (He et al., 2015)

Residual connections help mitigate the vanishing gradient problem and improve gradient flow in deep networks.

- A residual connection allows the gradient to bypass some layers, making training easier.
- Instead of learning the full transformation $F(x)$, the network learns the residual function $F(x) + x$.
- Some U-Net variants, such as ResUNet, integrate residual blocks for better performance.

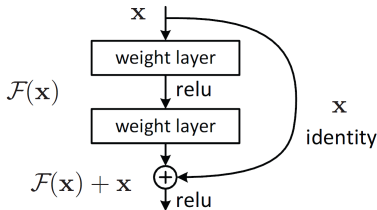


Figure: Residual connections in a CNN block.

Upsampling in U-Net

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- U-Net uses upsampling to restore spatial resolution in the decoder.
- Two main types of upsampling:
 - **Interpolation-based Upsampling:** Bilinear or nearest neighbor interpolation followed by convolution.
 - **Transposed Convolution (Deconvolution):** Learnable filters upscale feature maps.

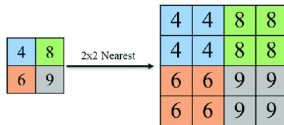


Figure: Nearest neighbor interpolation [Prince, 2023].

Transposed Convolution (Deconvolution)

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- Transposed convolution is a learnable upsampling method.
- Instead of direct interpolation, it applies convolution on an expanded input.
- Steps:
 - ① Insert zeros between pixels (if stride $\neq 1$).
 - ② Convolve with a learnable kernel.
 - ③ Output has a larger spatial resolution.

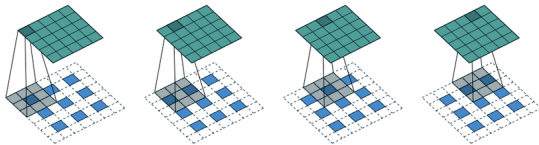


Figure: Example of transposed convolution operation.

U-Net: Encoder–Decoder with Skip Connections

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Main idea (U-Net)

Contracting path learns semantics; **expanding path** restores resolution. **Skip connections** fuse low-level spatial detail with high-level context.

Computation pattern (abstract)

Let E_ℓ be encoder features, D_ℓ decoder features:

$$E_{\ell+1} = \phi(\text{Down}(E_\ell)), \quad D_\ell = \phi(\text{Up}(D_{\ell+1}) \oplus E_\ell)$$

where $\oplus$ is concatenation (or sum), ϕ is Conv+Norm+Act. Final prediction:
 $P = \text{Softmax}(\text{Conv}_{1 \times 1}(D_0))$.

Implementation notes

- Up-sampling: bilinear+conv (stable) or transposed conv (learnable).
- Normalization: BatchNorm / GroupNorm (small batch) for segmentation.
- Deep supervision sometimes helps: auxiliary heads at intermediate scales.

U-Net: Training Algorithm

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Algorithm 1 U-Net Training Procedure

Require: Training dataset $\mathcal{D} = \{(x^{(n)}, y^{(n)})\}_{n=1}^M$

Require: Parameters θ , learning rate η

```
1: for epoch = 1, ..., E do
2:   for mini-batch  $\{(x_b, y_b)\}_{b=1}^B$  do
3:      $P_b \leftarrow f_{\theta}(x_b)$  ▷ Forward pass (encoder-decoder)
4:      $\mathcal{L}_{CE} \leftarrow \text{CrossEntropy}(P_b, y_b)$ 
5:      $\mathcal{L}_{Dice} \leftarrow \text{DiceLoss}(P_b, y_b)$ 
6:      $\mathcal{L} \leftarrow \lambda_{CE} \mathcal{L}_{CE} + \lambda_{Dice} \mathcal{L}_{Dice}$ 
7:      $\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}$  ▷ Backpropagation
8:   end for
9: end for
10: return trained parameters  $\theta$ 
```

Remarks

- Encoder may be pretrained (ImageNet initialization).
- Data augmentation applied jointly to (x, y) before forward pass.

DeepLab: Atrous Convolution and ASPP

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Main idea (DeepLab family)

Increase receptive field **without downsampling too much** using atrous (dilated) conv, then aggregate multi-scale context with **ASPP**.

Atrous (dilated) convolution

For 1D intuition (extends to 2D):

$$y[t] = \sum_k w[k] x[t + r \cdot k]$$

r is dilation rate ($r = 1$ is standard conv). Larger r increases receptive field.

ASPP (multi-scale context)

Parallel atrous conv branches with different dilation rates $\{r_1, r_2, r_3, \dots\}$:

$$\text{ASPP}(F) = \text{Concat}(\text{Conv}_{r_1}(F), \text{Conv}_{r_2}(F), \dots)$$

Training Setup: Data, Augmentations, and Splits

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Data formats

- Input: image $x \in \mathbb{R}^{H \times W \times 3}$
- Label map: $y \in \{1, \dots, K\}^{H \times W}$ (or ignore label 0 / 255)

Batching strategy (practical)

- Random crop to fixed size (e.g., 512×512), keep aspect via resize+pad.
- Balance rare classes: class-aware sampling or oversample patches containing target.

Augmentations (segmentation-safe)

- Geometric: flips, rotations, scale jitter, random crop (apply to image and mask).
- Photometric: brightness/contrast, blur/noise (image only).
- CutMix/MixUp variants exist but need care with masks (often helps with domain shift).

Losses for Semantic Segmentation

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Pixel-wise Cross-Entropy (multiclass)

Let $p_u(k) = P(u, k)$ and one-hot target $\mathbf{1}[y_u = k]$: $\mathcal{L}_{CE} = -\frac{1}{HW} \sum_u \sum_{k=1}^K \mathbf{1}[y_u = k] \log p_u(k)$

Class-weighted CE for imbalance: $\mathcal{L}_{WCE} = -\frac{1}{HW} \sum_u w_{y_u} \log p_u(y_u)$

Dice loss (binary; per-class for multiclass)

For a class k (soft prediction $\hat{y}_u = p_u(k)$, target $y_u \in \{0, 1\}$):

$$Dice_k = \frac{2 \sum_u y_u \hat{y}_u + \epsilon}{\sum_u y_u + \sum_u \hat{y}_u + \epsilon}, \quad \mathcal{L}_{Dice} = 1 - \frac{1}{K} \sum_{k=1}^K Dice_k$$

Common composite objective

$$\mathcal{L} = \lambda_{CE} \mathcal{L}_{(W)CE} + \lambda_{Dice} \mathcal{L}_{Dice}$$

Optimization Recipe (Baseline That Works)

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Baseline training loop (high-level)

- Backbone pretraining: ImageNet (common) or domain-specific (medical).
- Optimizer: AdamW (transformers) or SGD+momentum (CNNs).
- LR schedule: poly decay / cosine; warmup for transformers.
- Regularization: weight decay; dropout mainly in transformer decoders.

Typical hyperparameters (starting point)

- Crop: 512^2 (Cityscapes) / 256^2 – 512^2 (medical, depends on resolution)
- Batch: as large as fits; if small, prefer GroupNorm and gradient accumulation
- $\lambda_{CE} = 1$, $\lambda_{Dice} \in [0.5, 2]$ (tune per dataset imbalance)

Two common failure modes

- (1) Over-smoothing / blurry masks → increase decoder capacity or resolution.
- (2) Rare class collapse → re-weighting, patch sampling, Dice/Tversky, focal-like terms.

Evaluation: Metrics and What They Diagnose

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IoU and mIoU

For class k , with predicted set $\hat{Y}_k$ and ground-truth set Y_k :

$$IoU_k = \frac{|Y_k \cap \hat{Y}_k|}{|Y_k \cup \hat{Y}_k|}, \quad mIoU = \frac{1}{K} \sum_{k=1}^K IoU_k$$

Dice (F1 for masks)

$$Dice_k = \frac{2|Y_k \cap \hat{Y}_k|}{|Y_k| + |\hat{Y}_k|}$$

Dice is often more sensitive to **small structures** than IoU.

Qualitative checks (must-have)

Boundary quality, thin objects, small targets, failure under illumination / viewpoint change.

Analytics: Model Comparison and Selection Logic

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Comparative summary (semantic segmentation)

Family	Strength	Weakness	When to use	Notes
U-Net / Enc-Dec	strong locality, stable training	limited global context	small/medium data, medical	skip connections are key
DeepLab (ASPP)	multi-scale context	heavier, tuning needed	natural scenes, scale variance	good with strong backbone
Transformer (SegFormer)	global context, strong transfer	data hungry, compute	large data, domain shift	AdamW + warmup

Selection logic (engineering)

Real-time / embedded: lightweight encoder-decoder; highest accuracy: strong backbone + DeepLab/Transformer; small target / imbalance: Dice/Tversky + patch sampling.

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Section 4. Object Detection

Object Detection: Problem Setup

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Goal (set prediction)

Given image $x \in \mathbb{R}^{H \times W \times 3}$, predict a set of objects:

$$f_{\theta}(x) = \{(b_i, c_i, s_i)\}_{i=1}^N, \quad b_i = (x_i, y_i, w_i, h_i) \in \mathbb{R}^4,$$

$$c_i \in \{1, \dots, K\}, \quad s_i \in [0, 1]$$

Why it is hard

- Strong imbalance: **many negatives** vs few objects.
- Scale variation: small/large objects; occlusion; clutter.
- Joint **classification** + **localization**.



YOLO Labels: Matching Predictions to Ground Truth

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Training targets

Given GT objects $\{(b_j^*, c_j^*)\}_{j=1}^{N^*}$, assign each GT to one (or few) predictions. Modern YOLO variants use **dynamic assignment** (e.g., IoU + classification quality).

Simplified assignment view

- Define candidate predictions near the GT center and across scales.
- Compute quality $q(u)$ using IoU and/or classification logit.
- Select top- m candidates as positives; others negatives.

Practical note

Bad assignment $\Rightarrow$ unstable training (box regression fights classification). Most accuracy gains in modern detectors come from better **label assignment** + **loss design**.

YOLO Architecture

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- YOLO follows a **Backbone-Neck-Head** design.
- **Backbone**: extracts hierarchical features from the input image.
- **Neck (FPN/PAN)**: fuses multi-scale features for small and large object detection.
- **Detection Head**: predicts bounding boxes, objectness scores, and class probabilities.
- Final predictions are decoded and filtered using **Non-Maximum Suppression (NMS)**.

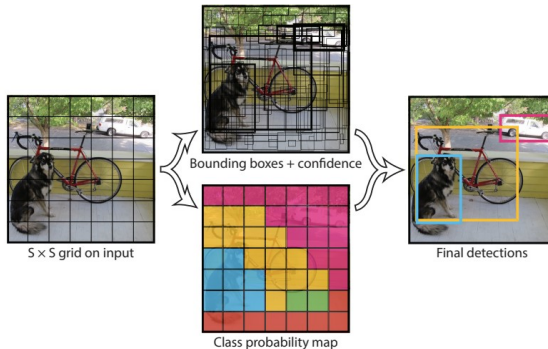


Figure: Typical YOLO architecture (Backbone-Neck-Head pipeline) Redmon et al. [2016].

YOLO Loss: Box + Objectness + Class

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Composite objective

$$\mathcal{L}_{det} = \lambda_{box} \mathcal{L}_{box} + \lambda_{obj} \mathcal{L}_{obj} + \lambda_{cls} \mathcal{L}_{cls}$$

Box regression (IoU family)

Using IoU-based losses improves localization:

$$\mathcal{L}_{box} = 1 - IoU(b, b^*) \quad \text{or} \quad 1 - GloU/DIoU/CIoU$$

Objectness + classification

- $\mathcal{L}_{obj}$: BCE on $p_{obj}(u)$ (positive for matched predictions, negative otherwise).
- $\mathcal{L}_{cls}$: BCE/CE on class distribution for positive predictions.
- Focal-like modulation is often used to down-weight easy negatives.

YOLO Inference: Decoding and NMS

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Decoding

Predicted offsets Δb are converted to image-space boxes b (scale/stride dependent). Compute score $s = p_{obj} \cdot \max_k p_{cls}(k)$ (typical).

Non-Maximum Suppression (NMS)

Given candidate boxes with scores, suppress duplicates:

- Sort boxes by score (descending).
- Keep highest-score box; remove boxes with $IoU > \tau$ to it.
- Repeat until list exhausted.

Engineering knobs

Confidence threshold τ_s and NMS IoU threshold τ control precision/recall trade-off.

YOLO: Training & Inference Algorithm

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Algorithm 2 YOLO Pipeline (Training + Inference)

Require: Dataset $\mathcal{D} = \{(x^{(n)}, \{(b_j^*, c_j^*)\})\}_{n=1}^M$, params θ , lr η

```
1: for epoch = 1, ..., E do
2:   for mini-batch  $\{(x, GT)\}$  do
3:      $Z \leftarrow f_\theta(x)$  ▷ Dense predictions at multi-scales
4:      $\mathcal{A} \leftarrow \text{Assign}(Z, GT)$  ▷ Match GT to predictions (positives)
5:      $\mathcal{L} \leftarrow \lambda_{\text{box}} \mathcal{L}_{\text{box}}(Z, GT, \mathcal{A}) + \lambda_{\text{obj}} \mathcal{L}_{\text{obj}}(Z, \mathcal{A}) + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}}(Z, GT, \mathcal{A})$ 
6:      $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$ 
7:   end for
8: end for
```

9: **Inference:** $Z \leftarrow f_\theta(x)$; decode boxes and scores; keep boxes with $s > \tau_s$; apply NMS with IoU threshold τ .

Other Detectors (Comparison Table)

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Where YOLO sits in the detector landscape

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Family	Paradigm	Key components	Post-proc	Typical use
YOLO	one-stage	FPN/PAN + dense head, dynamic assign	NMS	real-time, robotics
Faster R-CNN	two-stage	RPN proposals + ROI head	NMS	max accuracy (offline)
RetinaNet	one-stage	anchors + focal loss	NMS	strong baseline (imbalanced)
DETR-like	set prediction	transformer + Hungarian matching	no NMS	clean pipeline, research

Selection logic (practical)

Real-time / edge $\Rightarrow$ YOLO. Highest accuracy (when speed is secondary) $\Rightarrow$ Faster R-CNN / strong two-stage. Clean end-to-end (no NMS) and good scaling with data $\Rightarrow$ DETR-family.

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Section 5. Segment Anything Model (SAM)

What is SAM?

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Key Idea

The Segment Anything Model (SAM) is a foundation model for segmentation developed by Meta AI. It can segment any object in an image using different input prompts, making it highly versatile.

- Works with bounding boxes, clicks, or freeform masks as prompts.
- Trained on SA-1B dataset with over 1 billion masks.
- Enables zero-shot segmentation on unseen images.

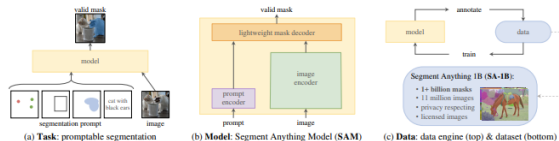


Figure: SAM Model Overview Kirillov et al. [2023].

SAM Architecture

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- **Image Encoder:** A large Vision Transformer (ViT) pre-trained using masked autoencoding.
- **Prompt Encoder:** Processes input prompts (points, boxes, masks).
- **Mask Decoder:** Predicts segmentation masks based on encoded image and prompts.

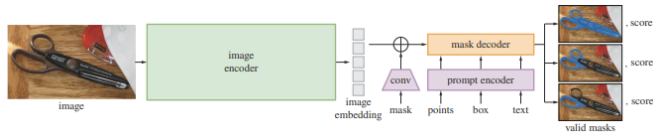


Figure: Segment Anything Model (SAM) Architecture.

Training and Dataset

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- **SA-1B Dataset:** The largest segmentation dataset with 1 billion masks.
- **Semi-Automatic Annotation:** Human annotators refined AI-generated masks.
- **Generalization:** Trained to segment unseen objects and domains.

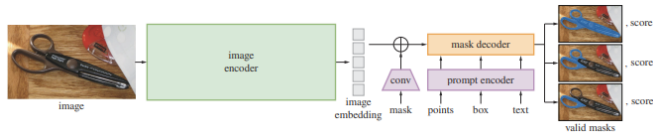


Figure: Segment Anything Model (SAM) Architecture.

How SAM Works?

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- The user provides an input prompt (point, bounding box, or mask).
- SAM generates segmentation masks based on prompt and image encoding.
- Output can be refined interactively.

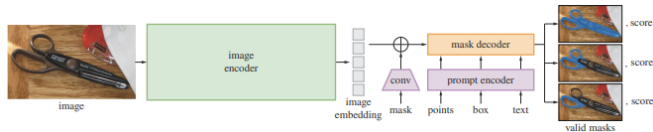


Figure: Segment Anything Model (SAM) Architecture.

Applications of SAM

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- **Medical Imaging:** Tumor and organ segmentation.
- **Autonomous Driving:** Road scene segmentation.
- **AR/VR:** Object segmentation for augmented reality.
- **Robotics:** Object recognition and manipulation.

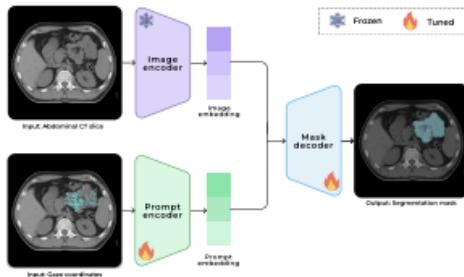


Figure: Medical Application of SAM.

Key Takeaways

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- SAM is a powerful, zero-shot segmentation model.
- Works with multiple input prompts for flexible segmentation.
- Trained on the largest segmentation dataset, enabling generalization.
- Open-source and widely applicable across industries.

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Section 6. Conclusion

Key Takeaways

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Conclusion

- Segmentation and detection are essential for computer vision applications.
- Deep learning-based models have achieved state-of-the-art performance.
- Practical implementation involves dataset preparation, model training, and evaluation.

Bibliography

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